Topology Architecture Questions

10 questions | Original English questions and source answers

Question 1

Which purpose does a northbound API serve in a controller-based networking architecture?

Options

A. facilitates communication between the controller and the applications

B. reports device errors to a controller

C. generates statistics for network hardware and traffic

D. communicates between the controller and the physical network hardware

Answer

A. facilitates communication between the controller and the applications

Source: CCNA 200-301 Questions.html | Question 1

Question 2

What is a characteristic of spine-and-leaf architecture?

Options

A. It provides greater predictability on STP blocked ports

B. It provides variable latency

C. Each device is separated by the same number of hops

D. Each link between leaf switches allows for higher bandwidth

Answer

C. Each device is separated by the same number of hops

Source explanation

Spine-leaf architecture is typically deployed as two layers: spines (such as an aggregation layer), and leaves (such as an access layer). Spine-leaf topologies provide high-bandwidth, low-latency (-> Answer 'It provides variable latency' is not correct), nonblocking server-to-server connectivity.

With a spine-and-leaf architecture, no matter which leaf switch to which a server is connected, its traffic always has to cross the same number of devices to get to another server (unless the other server is located on the same leaf) (-> Answer 'Each device is separated by the same number of hops' is correct) . This approach keeps latency at a predictable level because a payload only has to hop to a spine switch and another leaf switch to reach its destination.

Reference: https://www.cisco.com/c/en/us/products/collateral/switches/nexus-7000-series-switches/white-paper-c11-737022.html

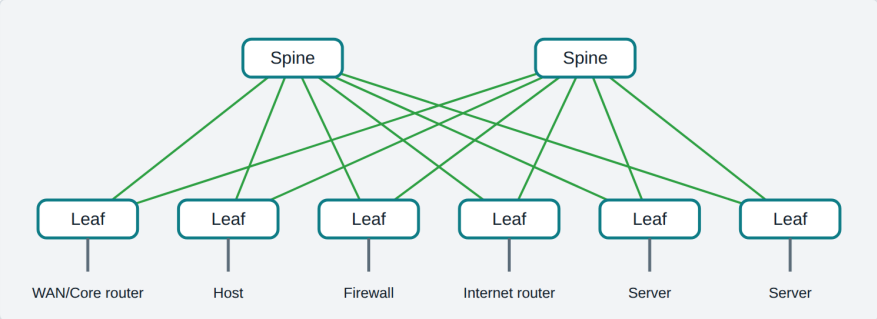
With Leaf-Spine, the network uses Layer 3 routing so STP is no longer required. (-> Answer 'It provides greater predictability on STP blocked ports' is not correct). Spine-leaf architectures rely on protocols such as Equal-Cost Multipath (ECPM) routing to load balance traffic across all available paths while still preventing network loops. This allows all connections to be utilized at the same time while still remaining stable and avoiding loops within the network.
SpineSpineLeafWAN/Core routerLeafHostLeafFirewallLeafInternet routerLeafServerLeafServer

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Source: CCNA 200-301 Questions.html | Question 2

Question 3

Which two functions are performed by the core layer in a three-tier architecture? (Choose two)

Options

A. Inspect packets for malicious activity

B. Provide uninterrupted forwarding service

C. Ensure timely data transfer between layers

D. Police traffic that is sent to the edge of the network

E. Provide direct connectivity for end user devices

Answer

B. Provide uninterrupted forwarding service

C. Ensure timely data transfer between layers

Source explanation

Considerations at the core layer include
+ Providing high-speed switching (i.e., fast transport)
+ Providing reliability and fault tolerance
+ Scaling by using faster, and not more, equipment
+ Avoiding CPU-intensive packet manipulation caused by security, inspection, quality of service (QoS) classification, or other processes

Reference: https://www.ciscopress.com/articles/article.asp?p=2202410&seqNum=4

Source: CCNA 200-301 Questions.html | Question 3

Question 4

A company needs to interconnect several branch offices across a metropolitan area. The network engineer is seeking a solution that provides high-speed converged traffic, including voice, video, and data on the same network infrastructure. The company also wants easy integration to their existing LAN infrastructure in their office locations. Which technology should be recommended?

Options

A. ISDN

B. Frame Relay

C. VSAT

D. Ethernet WAN

Answer

D. Ethernet WAN

Source explanation

Ethernet WAN offer the ability of using Ethernet over long-distance links.

Source: CCNA 200-301 Questions.html | Question 4

Question 5

What is the name of the layer in the Cisco borderless switched network design that is considered to be the backbone used for high-speed connectivity and fault isolation?

Options

A. access

B. network

C. data link

D. core

E. network access

Answer

D. core

Source explanation

The key layers are access, distribution, and core. The core layer is the network backbone that hierarchically connects several layers of the network design, providing for connectivity between end devices, computing and data storage services located within the data center and other areas, and services within the network.

Source: CCNA 200-301 Questions.html | Question 5

Question 6

Which two WAN architecture options help a business improve scalability and reliability for the network? (Choose two)

Options

A. dual-homed branches

B. single-homed branches

C. asynchronous routing

D. dynamic routing

E. static routing

Answer

A. dual-homed branches

D. dynamic routing

Source explanation

Dual-homed branches (branches connects to two ISPs or one ISP with two connections) increases reliability for the network.

Dynamic routing automatically use backup routes in the case of main routes fail which increases reliability. The configuration is remain unchanged or change a little when we expand the network thus increasing the scalability.

Note: A scalable network can expand quickly to support new users and applications without impacting the performance of the service being delivered to existing users

Source: CCNA 200-301 Questions.html | Question 6

Question 7

What are two descriptions of three-tier network topologies? (Choose two)

Options

A. The access layer manages routing between devices in different domains

B. The core and distribution layers perform the same functions

C. The core layer maintains wired connections for each host

D. The network core is designed to maintain continuous connectivity when devices fail

E. The distribution layer runs Layer 2 and Layer 3 technologies

Answer

D. The network core is designed to maintain continuous connectivity when devices fail

E. The distribution layer runs Layer 2 and Layer 3 technologies

Source: CCNA 200-301 Questions.html | Question 7

Question 8

What benefit does controller-based networking provide versus traditional networking?

Options

A. allows configuration and monitoring of the network from one centralized port

B. moves from a two-tier to a three-tier network architecture to provide maximum redundancy

C. provides an added layer of security to protect from DDoS attacks

D. combines control and data plane functionality on a single device to minimize latency

Answer

A. allows configuration and monitoring of the network from one centralized port

Source: CCNA 200-301 Questions.html | Question 8

Question 9

Which function is performed by the collapsed core layer in a two-tier architecture?

Options

A. attaching users to the edge of the network

B. applying security policies

C. marking interesting traffic for data polices

D. enforcing routing policies

Answer

D. enforcing routing policies

Source explanation

The three-tier hierarchical design maximizes performance, network availability, and the ability to scale the network design.
However, many small enterprise networks do not grow significantly larger over time. Therefore, a two-tier hierarchical design where the core and distribution layers are collapsed into one layer is often more practical. A “collapsed core” is when the distribution layer and core layer functions are implemented by a single device. The primary motivation for the collapsed core design is reducing network cost, while maintaining most of the benefits of the three-tier hierarchical model.

Reference: https://www.ciscopress.com/articles/article.asp?p=2202410&seqNum=4

A collapsed core network is shown below. The collapsed core network may be deployed with redundant core/distribution router, or consolidated core/distribution router.

Deploying a collapsed core network results in the distribution layer and core layer functions being implemented in a single device. The collapsed core/distribution device must provide the following:
+ High speed physical and logical paths connecting to the network
+ Layer-2 aggregation and demarcation point
+ Define routing and network access policies -> Answer 'enforcing routing policies' is correct
+ Intelligent network services—QoS, Network virtualization, etc.

Reference:
https://www.cisco.com/c/en/us/td/docs/solutions/Enterprise/Small_Enterprise_Design_Profile/SEDP/chap2.html

Source: CCNA 200-301 Questions.html | Question 9

Question 10

Which WAN access technology is preferred for a small office / home office architecture?

Options

A. Integrated Services Digital Network switching

B. frame-relay packet switching

C. dedicated point-to-point leased line

D. broadband cable access

Answer

D. broadband cable access

Source: CCNA 200-301 Questions.html | Question 10